

TrES-1b

R-0177

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_180423 · created 2026-07-16T18:41:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458231.8708 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1403 +/- 0.0083
Transit depth (Rp/Rs)^2	1.97 +/- 0.23 [%]
Orbital Inclination (inc)	88.18 +/- 0.79
Airmass coefficient 1 (a1)	1.03 +/- 0.011
Airmass coefficient 2 (a2)	-0.0218 +/- 0.0035
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	#2 - [233, 167]
Optimal Aperture	4.02
Optimal Annulus	11.85
Transit Duration (day)	0.0991 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1403 ± 0.0083 vs 0.1358 (deviation 0.54σ)
Duration fit vs published	0.0991 d vs 0.1042 d (deviation 1.13σ)
Mid-time O–C	-4.0 min
Depth SNR	16.9
Residual scatter	1.05 %

Light curve

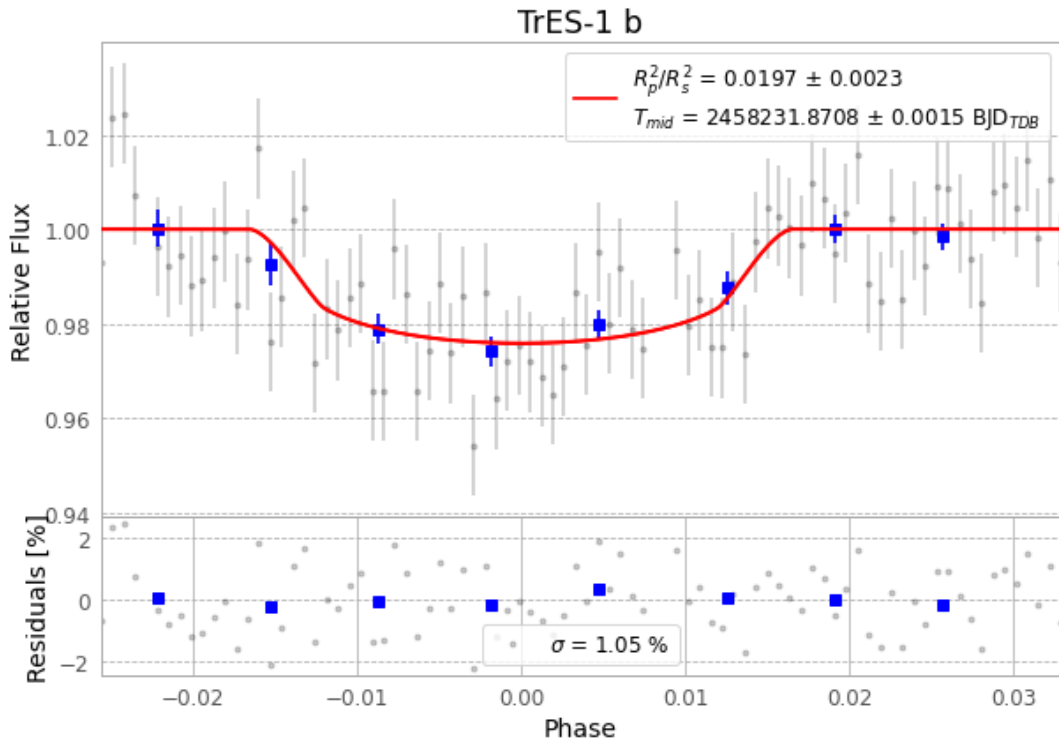


Figure 1 — FinalLightCurve_TrES-1 b_2018-04-23.png (SHA-256 fd436f4bbdf1c741...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 168], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3ecf95ba65182dee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f67d25a

Data provenance

84 FITS frames (55 MB), TRES-1180423070012.FITS → TRES-1180423111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-180423.log (0d5c2161785b...), FinalLightCurve_TrES-1 b_2018-04-23.png (fd436f4bbdf1...), FinalLightCurve_TrES-1 b_2018-04-23.csv (7da9f1417719...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 18:41Z.